

Requests ①
TTFT / TPOT SLO (ms)



Chatbot
100 / 150



Summary
800 / 100



Translation
3000 / 30

② **Deadline-Aware Request Dispatcher**

Prefill Request
Arrival + TTFT SLO

Decode Request
Last token + TPOT SLO

Waiting

R₄ R₅ R₃ ...

*Phase-Aware
Request Ordering*

*Deadline-Aware
Preemption*

Running

R₄ R₁ R₂ R₃ ...

③ **Step-Time-Model-Guided Token Allocator**

Prefill Decode
KV context Attention

Decode deadlines
D₁ D₂ D₃ ...

Step-time model ⌚

Predicted time

Time budget

Token Allocation

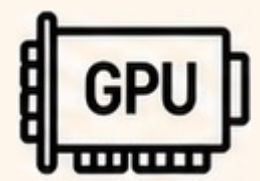
Prefill Decode Blank

TPOT budget *Token cap*



Better Multi-SLO Goodput

④



vLLM Execution
Continuous batching